ADASYN-GRU Approach for Enhanced Detection of Heavy Metals in Water Using Electrochemical Sensors

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***Abstract*—Detecting heavy metals in water is critical for environmental monitoring. Electrochemical sensors provide a reliable solution that is easy to use, simple, fast, and cost-effective. This study presents an Adaptive Synthetic Sampling Approach for Imbalanced Learning-Gated Recurrent Unit (ADASYN-GRU) approach to improve the accuracy of heavy metal detection in water samples. The dataset consists of 606 samples distributed across three classes: Cadmium (Cd), Non-Heavy Metal, and Lead (Pb). To address class imbalance, data balancing techniques such as Synthetic Minority Over-sampling Technique (SMOTE), Generative Adversarial Network (GAN), and Adaptive Synthetic Sampling Approach for Imbalanced Learning (ADASYN) were applied. Five deep learning models, including Artificial Neural Network (ANN), Deep Neural Network (DNN), Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), and Gated Recurrent Unit (GRU), were evaluated using stratified 5-fold cross-validation. The combination of ADASYN-GRU achieved the highest performance, with an accuracy of 96.54%.**

Keywords—ADASYN, deep learning, electrochemical sensors, GRU, heavy metal.

I. INTRODUCTION

The presence of heavy metals in water sources has become a significant environmental concern, primarily due to their detrimental effects on the surrounding ecosystem [1]. This contamination is attributable to a number of factors, including industrial waste, mining activities, and the use of chemicals in the agricultural sector [2, 3]. The contamination of water with heavy metals not only has an adverse impact on the ecosystem but also poses a significant risk to human health when used for routine activities such as drinking, cooking, and bathing. Exposure to heavy metals has been linked to a range of adverse health effects, including nervous system disorders, kidney damage, and an increased risk of cancer disease [4, 5].

The current methods used to detect heavy metals typically employ laboratory techniques, including atomic absorption spectroscopy (AAS), inductive plasma mass spectrometry (ICP-MS), and ion chromatography [6–8]. These methods are known for their high sensitivity, enabling the detection of heavy metals in very low concentrations. However, these techniques are not without their limitations. The equipment and chemicals utilized are prohibitively expensive, rendering them inaccessible to a significant proportion of the population [7]. Furthermore, this method necessitates the involvement of trained operators and is associated with relatively prolonged analysis times, which renders it less efficacious for real-time monitoring applications. Conversely, the considerable energy consumption and its environmental impact represent additional considerations [8].

Electrochemical sensors have emerged as a highly effective alternative to conventional heavy metal detection methods, such as atomic absorption spectroscopy (AAS) and inductively coupled plasma mass spectrometry (ICP-MS). The principal benefit of these sensors is their portability, which permits direct utilization in the field without the necessity of transporting samples to a laboratory setting[9]. Furthermore, electrochemical sensors have lower operational costs, as they do not require complex equipment or expensive consumables [10]. The brief analysis time is also advantageous, enabling real-time measurements with rapid results [11]. With high sensitivity and the capacity to detect heavy metals in low concentrations, these sensors provide an efficient and environmentally friendly solution for water quality monitoring.

The application of deep learning techniques to complement electrochemical sensors in predicting heavy metal concentrations is a critical issue, given the complex data sets generated by such sensors. Electrochemical data are often non-linear and exhibit hidden patterns that are difficult to interpret using traditional analytical methods [12]. Deep learning offers the ability to process large data sets with high precision, facilitating the identification and prediction of metal concentrations with greater accuracy[13]. In addition, deep learning algorithms can investigate the intricate relationships between electrochemical signals and metal concentrations, thereby improving the sensitivity and accuracy of the detection system [14]. By integrating deep learning, electrochemical sensors can provide more reliable real-time predictions for water quality monitoring..

Balancing or resampling is an important step in data analysis for heavy metal detection using electrochemical sensors and deep learning methods. The data generated from sensor measurements is often unbalanced, with the number of samples with a certain heavy metal concentration much lower than other categories. This imbalance can lead to bias in prediction models, where algorithms tend to be more accurate for dominant classes but less accurate for minority classes [15]. Resampling, such as oversampling the minority class or undersampling the majority class, helps to create a more balanced data distribution so that the model can learn fairly from all categories [16] By balancing, deep learning models can improve overall accuracy and sensitivity to lower concentrations of heavy metals, which is crucial for early detection and control of pollution.

Several previous studies have developed electrochemical sensors for the detection of heavy metals. For example, [3]fabricated and characterized electrochemical sensor electrodes based on thick-film technology to detect cadmium. However, this study did not mention the amount of data used, the level of accuracy achieved, or whether or not data balancing was performed. [17] used a screen-printed carbon electrode modified with bismuth for cadmium analysis using cyclic voltammetry. This study also did not provide information on the amount of data, accuracy and use of data balancing. [18, 19]. Both studies show the potential of electrochemical sensors in the detection of heavy metals, but the lack of information on data volume, accuracy, and data balancing techniques indicates a gap that needs to be filled. Further research is needed to integrate machine learning techniques, such as deep learning, which require large and balanced data sets to improve prediction accuracy.

The purpose of this research is to explore the most superior deep learning methods in predicting output data from electrochemical sensors, using various algorithms such as Artificial Neural Network (ANN), Deep Neural Network (DNN), Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), and Gated Recurrent Unit (GRU) [20, 21]. In addition, this research also aims to identify the most effective data balancing method to improve the accuracy of deep learning models through the application of various resampling techniques, such as Synthetic Minority Oversampling Technique (SMOTE), Generative Adversarial Networks (GAN), and Adaptive Synthetic Sampling (ADASYN) [22]. The combination of resampling methods and deep learning algorithms is expected to provide more accurate and reliable prediction results.

The purpose of this study is to compare compensation methods to improve the accuracy of detecting fallow metals in water using electrochemistry. There are several sections in this paper: Section 2 presents related research. Section 3 details the methods used in this study. Section 4 describes the results of the study. The last section provides the conclusion of this paper.

II. Proposed Method

This research begins with the sample preparation stage, where data is collected from the output of electrochemical sensors to detect heavy metals. The data obtained is then subjected to a balancing process using methods such as SMOTE, GAN, or ADASYN to address the imbalance in data distribution. Next, dimensionality reduction is performed using Principal Component Analysis (PCA) to simplify data complexity without losing important information. The reduced data is then classified using deep learning algorithms, such as ANN, DNN, LSTM, RNN, or GRU. Finally, the resulting models are evaluated using accuracy, metric to ensure optimal performance. The proposed method in this research is illustrated in Figure 1.

A. Data Collection using Electrochemical Sensor

Electrochemical sensors are widely utilized for detecting various substances, including heavy metals in water. These sensors operate by measuring the voltage (V) and current (μA) generated during electrochemical reactions. In this study, the output of the electrochemical sensor consists of two key parameters: voltage and current. Each sample contains approximately 800 rows for each parameter, which are then flattened to create a single feature vector of 1600 features per sample.

The dataset comprises 606 samples, categorized into three classes: Cadmium (Cd), Non-Heavy Metal, and Lead (Pb). The distribution of samples across these classes is as follows: 195 samples for Cadmium, 93 samples for Non-Heavy Metal, and 318 samples for Lead.

Given the high dimensionality of the dataset, dimensionality reduction is critical to address computational inefficiencies and mitigate the risk of overfitting. Principal Component Analysis (PCA) is employed as the primary dimensionality reduction technique. PCA transforms the original high-dimensional dataset into a smaller set of features while retaining most of the variance, which is essential for preserving meaningful information.

B. Balancing Methods

There are various balancing methods, including SMOTE, GAN, and ADASYN. SMOTE (Synthetic Minority Oversampling Technique), GAN (Generative Adversarial Networks), and ADASYN (Adaptive Synthetic Sampling) are methods used to address data imbalance in machine learning. SMOTE works by creating synthetic samples of the minority class through interpolation between the original data, thus improving the class distribution without duplicating the data [23]. For SMOTE, the synthetic sample (x_{new}) is generated using Eq. (1).

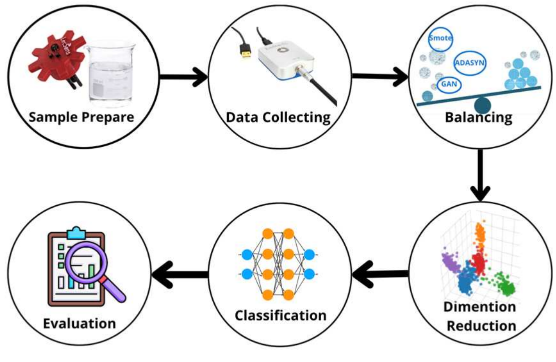

Fig. 1. The proposed method

$$x_{new} = x_i + \lambda \cdot (x_j - x_i) \quad (1)$$

where x_i is the minority class sample, x_j is its nearest neighbor, and λ is a random number between 0 and 1. This approach helps to balance the dataset by generating synthetic examples in a similar feature space to the original data.

GAN, on the other hand, uses two neural networks, a generator and a discriminator, to generate synthetic data that resembles the original data with high-quality [24]. Meanwhile, ADASYN is a more adaptive development of SMOTE, where synthetic samples are focused on regions of the data that are difficult to separate, helping the model better understand complex patterns in minority classes [25, 26].

These three methods complement each other in handling data imbalance. SMOTE is suitable for simple scenarios with relatively homogeneous distributions, while GAN is more effective for generating complex and realistic data. ADASYN offers an adaptive approach that considers data regions that are difficult to classify, providing more flexibility in certain situations.

C. Deep Learning Models

There are various deep-learning methods for classification tasks, including ANN, DNN, LSTM, RNN, and GRU. Artificial Neural Network (ANN) is a basic model in deep learning that mimics the way the human brain works by utilizing interconnected artificial neurons [27]. ANNs consist of input layers, hidden layers, and output layers, and can be used for tasks such as classification and regression. The forward pass in an ANN is calculated using Eq. (2).

$$\boldsymbol{y} = \boldsymbol{f}\left(\sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{w_i x_i} + \boldsymbol{b}\right) \quad (2)$$

where $\boldsymbol{y}$ is the output, $\boldsymbol{w_i}$ are the weights, $\boldsymbol{x_i}$ are the inputs, $\boldsymbol{b}$ is the bias, and $\boldsymbol{f}$ is the activation function (e.g., sigmoid, ReLU).

ANNs form the basis for the development of more complex models, such as Deep Neural Networks (DNN). DNN is an advanced form of ANN that has many hidden layers, enabling more complex data modeling with deep feature abstraction [28]. DNNs are often used in image and sound processing due to their ability to understand complex patterns, although they require greater computational resources.

Recurrent Neural Network (RNN) extends the capabilities of ANN by introducing feedback connections, which enables the processing of sequential data such as text, voice, and time series [29]. However, RNN has limitations in handling long data due to the vanishing gradient problem, where important information from the beginning of the sequence can be lost over time. To overcome this drawback, Long Short-Term Memory (LSTM) was developed as a type of RNN that uses a gating mechanism to retain information over a longer period of time [30]. LSTMs are very effective for handling complex temporal relationships, such as sentiment analysis in text or time series prediction.

Gated Recurrent Unit (GRU) is a variant of LSTM that simplifies the architecture by using two main gates, namely reset and update gates [31]. This makes GRU computationally lighter than LSTM, while still maintaining the ability to handle sequential data. GRU is often chosen when computational efficiency is a priority without significant performance loss.

D. Evaluation Metrics

To measure the performance of the proposed methods, accuracy is used as the primary indicator. Accuracy represents the proportion of correctly predicted data to the total amount of data, as shown in Eq. (3). In this equation, TP represents the number of data points correctly classified as positive, TN is the number of data points correctly classified as negative, FP refers to data points incorrectly classified as positive, and FN refers to data points incorrectly classified as negative.

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \quad (3)$$

III. Experimental Result

A. Result of Balancing Methods

Balancing the dataset was an essential step to address the significant class imbalance in the raw data. The initial distributions showed 195 data points for Cadmium (Cd), 93 for Non-Heavy Metal, and 318 for Lead (Pb). This imbalance could lead to biased classification results and reduced model performance. To resolve this issue, three balancing methods were applied: SMOTE, GAN, and ADASYN. The unbalanced dataset was also used as a baseline for comparison.

Each method successfully improved the class distributions. SMOTE was implemented using the imbalanced-learn library. SMOTE balanced all classes to 318 data points, ensuring equal representation for each class. GAN was implemented using TensorFlow. This method produced the same result as SMOTE, balancing the dataset to 318 data points for Cadmium, Non-Heavy Metal, and Lead. ADASYN, also implemented using the imbalanced-learn library, created a slightly different distribution. ADASYN generated 325 data points for Cadmium, 315 for Non-Heavy Metal, and 318 for Lead. These variations occurred because ADASYN focuses on generating data for more challenging samples within the minority class.

The results of the balancing methods are visualized in Figures 2, 3, and 4. Figure 2 shows the class distributions before and after balancing using SMOTE. Figure 3 presents the class distributions before and after balancing using GAN. Figure 4 illustrates the class distributions before and after balancing using ADASYN.

B. Result of Dimension Reduction

Dimensionality reduction was an essential step in this study due to the high-dimensional nature of the dataset, which contained 1600 features per data point. Such a large number of features increased computational demands and posed a higher risk of overfitting. To address these issues, Principal Component Analysis (PCA) was used to reduce the dimensionality of the dataset while preserving most of its variance.

The PCA process began with normalizing the dataset using StandardScaler. This normalization ensured that all features had a mean of 0 and a standard deviation of 1, a necessary step because PCA is sensitive to the scale of input features. After normalization, PCA identified principal components by computing the eigenvalues and eigenvectors of the covariance matrix. These principal components represent the directions in the data that explain the highest variance.

The optimal number of principal components depended on the balancing method applied. For the unbalanced dataset and those balanced using SMOTE or ADASYN, 14 principal components retained at least 98% of the dataset's variance. However, the dataset balanced with GAN required 44 components to achieve the same variance threshold.

This difference can be attributed to how each balancing method generates synthetic data. GANs create new samples by learning the distribution of the original dataset through adversarial training, which often introduces additional variability or noise. This added variability increases the total variance in the dataset, requiring more principal components to retain the same proportion of variance. In contrast, SMOTE and ADASYN generate synthetic samples by interpolating between existing data points, which has a more controlled effect on the variance structure. As a result, fewer components are needed to capture most of the variance when these methods are used.

C. Result of Classification Process

The classification process in this study aimed to evaluate the effectiveness of deep learning models combined with various data-balancing methods for detecting heavy metals in water. Five deep-learning approaches were tested: ANN, DNN, LSTM, RNN, and GRU. Stratified K-Fold Cross-Validation with k=5 was used to ensure fair evaluation. This method maintains the class proportions in each fold, reducing bias toward the majority class and providing more reliable results.

The performance of each model, measured using accuracy, is summarized in Table I. Models trained on the imbalanced dataset consistently showed suboptimal performance, with none achieving an accuracy exceeding 95%. This result underscores the significance of applying balancing techniques, as imbalanced data tends to bias models towards the majority class, leading to skewed predictions. Among the models trained on imbalanced data, GRU demonstrated the lowest accuracy at 94.56%, reflecting its struggle to effectively generalize under such conditions.

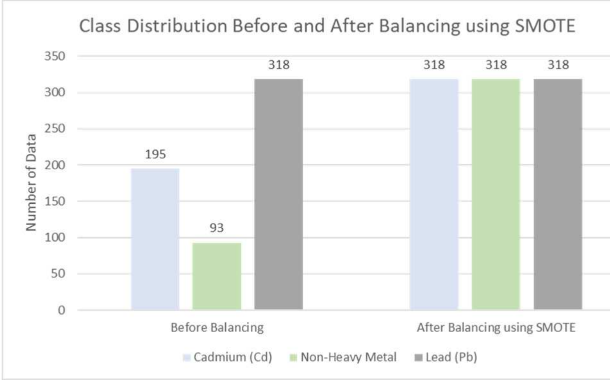

Fig. 2. Class distribution before and after balancing using SMOTE

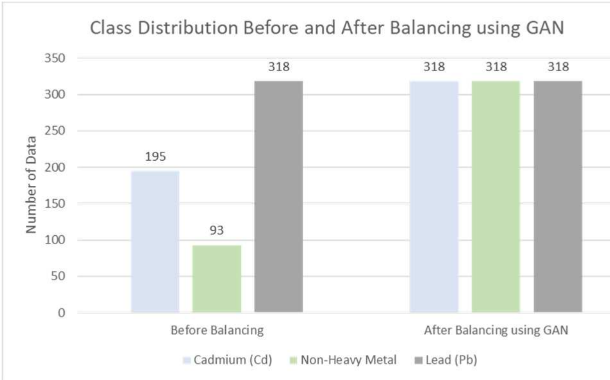

Fig. 3. Class distribution before and after balancing using GAN

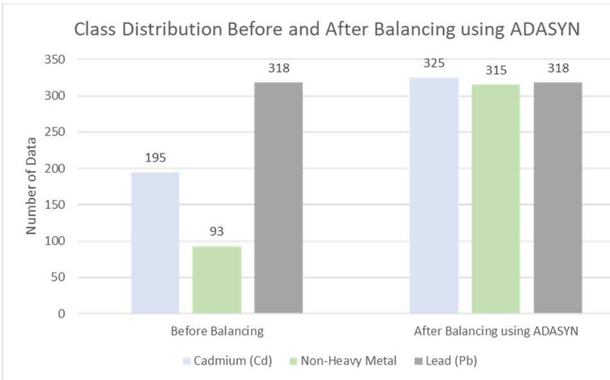

Fig. 4. Class distribution before and after balancing using ADASYN

TABLE I. THE ACCURACY OF EACH METHOD

Classifier	***Acc***			
	Non-Balancing	***SMOTE***	***GAN***	***ADASYN***
ANN	95.38	89.44	88.28	96.04
DNN	95.05	89.44	88.93	96.53
LSTM	94.89	81.68	82.51	89.45
RNN	94.72	86.15	88.12	95.55
GRU	94.56	88.12	88.12	**96.54**

Notes : All values are expressed as percentages (%). Accuracy is denoted as "***Acc***"

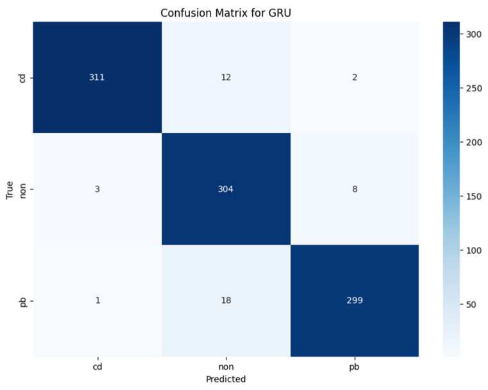

Fig. 5. Confusion matrix of ADASYN-GRU

However, the application of balancing techniques did not always lead to better performance. While some models trained on balanced datasets demonstrated significant accuracy improvements, others experienced notable declines, with several accuracies falling below 90%. For instance, SMOTE and GAN often resulted in degraded performance compared to the imbalanced dataset. SMOTE-GRU and GAN-GRU achieved accuracies of 88.12%, which are considerably lower than the baseline GRU accuracy of 94.56%. This decline highlights the limitations of these methods. SMOTE's interpolation-based approach creates synthetic samples by averaging neighboring data points, which may fail to represent the true distribution in complex datasets. Meanwhile, GAN relies on adversarial training to generate synthetic samples, but suboptimal training or imbalanced generator-discriminator dynamics may lead to unrealistic or noisy data that hinders model performance.

Among the balancing techniques, ADASYN emerged as the most effective. Its adaptive sampling strategy focuses on generating synthetic samples in regions of higher complexity, helping models to better understand minority class patterns. The ADASYN-GRU combination achieved the highest accuracy of 96.54%, outperforming all other methods. This superior performance can be attributed to the synergy between ADASYN and GRU. ADASYN effectively enhances dataset balance and diversity, while GRU, known for capturing sequential dependencies, fully leverages this enriched dataset to identify intricate patterns in the electrochemical data.

Other models paired with ADASYN also demonstrated notable performance improvements, such as ADASYN-ANN (96.04%) and ADASYN-DNN (96.53%). However, none surpassed the ADASYN-GRU combination, indicating that GRU's architecture is particularly well-suited to capitalize on ADASYN's adaptively generated samples. In contrast, alternative pairings such as SMOTE-GRU and GAN-GRU failed to achieve comparable results, further reinforcing the importance of aligning balancing strategies with the underlying model architecture.

The confusion matrix for the ADASYN-GRU combination is presented in Figure 5, highlighting the model's exceptional ability to classify all classes accurately with minimal errors. Additionally, the accuracy of all combinations is visualized in Figure 6, emphasizing the superiority of ADASYN-GRU compared to other methods.

IV. Conclusion

This study highlights the critical role of addressing class imbalance in improving the detection of heavy metals in water using electrochemical sensors. Balancing methods such as SMOTE, GAN, and ADASYN significantly enhanced model performance, with ADASYN emerging as the most effective due to its adaptive sampling strategy. The combination of ADASYN and GRU achieved the highest performance, with an accuracy of 96.54%. This superior result demonstrates the synergy between ADASYN's ability to generate high-quality synthetic samples and GRU's capability to capture sequential patterns in the electrochemical data. Future research could explore advanced feature selection techniques, validate the ADASYN-GRU approach on other datasets, and investigate novel balancing methods to further enhance classification performance.

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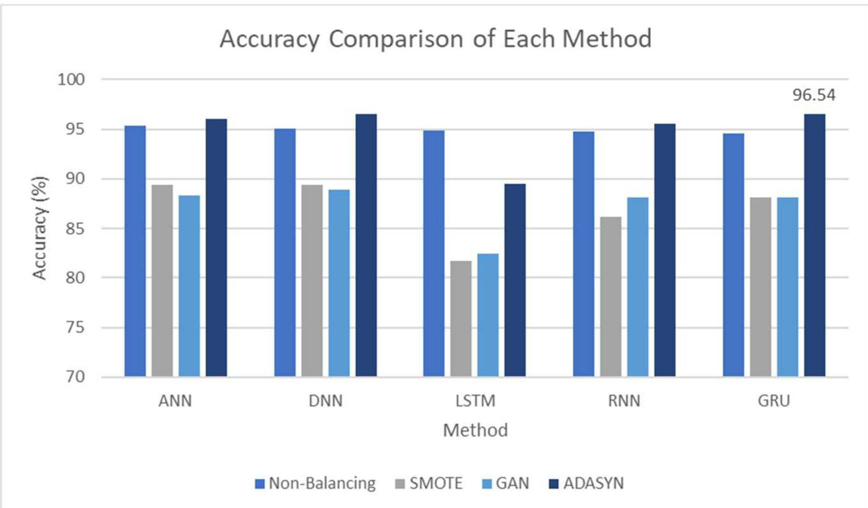

Fig.6. Accuracy Comparison of Each Method

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